



Teacher's Name _____

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Student Number

Knox Grammar School

2013

**Trial Higher School Certificate
Examination**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time - 3 hours
- Write using blue or black pen only
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Subject Teachers

Mr I Bradford
Mr M Vuletich

Setter

Mr Vuletich

**This paper MUST NOT be removed from the
examination room**

Number of Students in Course: 31

Total Marks – 100

Section I 10 Marks

- Answer Questions 1 to 10
- Use the Multiple Choice Answer Sheet

Section II 90 Marks

- Answer Questions 11 to 16
- All questions are worth 15 marks
- Answer each question in a separate Writing Booklet.

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Section I

10 Marks

Attempt Questions 1–10

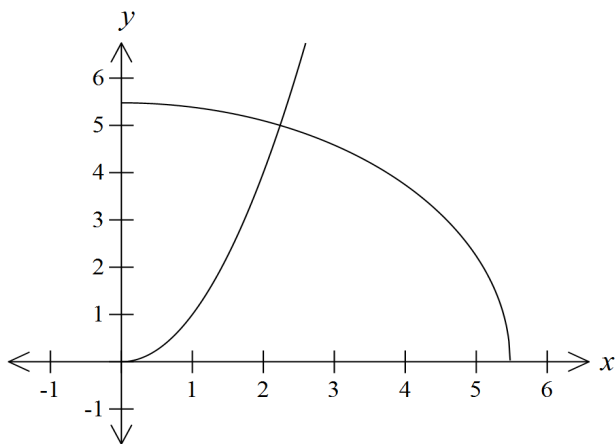
Allow about 20 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1 Let $z = 1 + 2i$ and $w = -2 + i$. What is the value of $\frac{5}{iw}$?

- (A) $-1 - 2i$
- (B) $-1 + 2i$
- (C) $1 - 2i$
- (D) $1 + 2i$

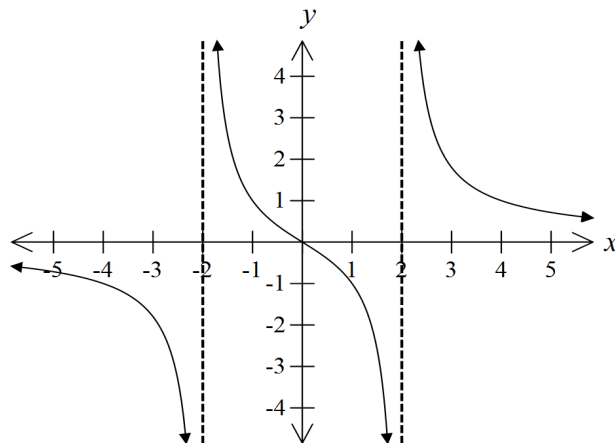
2 What is the volume of the solid formed when the region bounded by the curves, $y = x^2$, $y = \sqrt{30 - x^2}$ and the y-axis is rotated about the y-axis?



What is the correct expression for volume of this solid using the method of cylindrical shells?

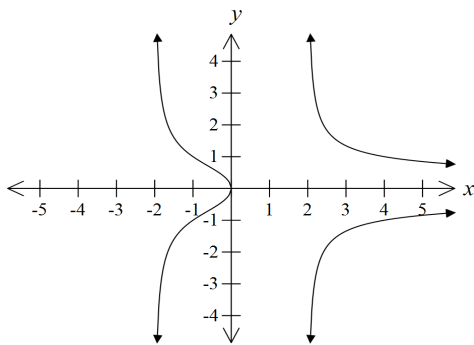
- (A) $V = \int_0^{\sqrt{5}} 2\pi \left(x^2 - \sqrt{30 - x^2} \right) dx$
- (B) $V = \int_0^{\sqrt{5}} 2\pi x \left(x^2 - \sqrt{30 - x^2} \right) dx$
- (C) $V = \int_0^{\sqrt{5}} 2\pi \left(\sqrt{30 - x^2} - x^2 \right) dx$
- (D) $V = \int_0^{\sqrt{5}} 2\pi x \left(\sqrt{30 - x^2} - x^2 \right) dx$

3 The diagram shows the graph of the function $y = f(x)$.

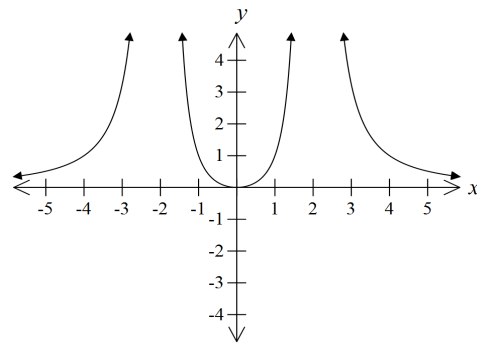


Which of the following is the graph of $y^2 = f(x)$?

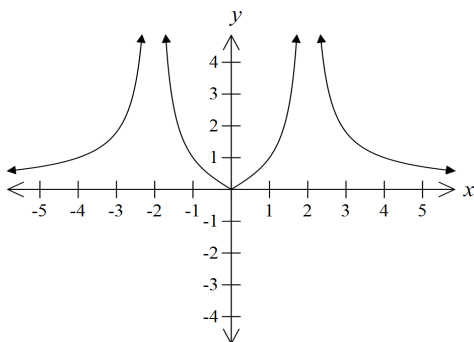
(A)



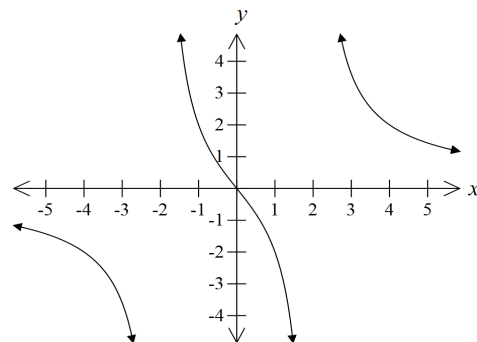
(B)



(C)



(D)



- 4 Let α, β and γ be roots of the equation $x^3 + 3x^2 + 4 = 0$. Which of the following polynomial equations have roots α^2, β^2 and γ^2 ?

- (A) $x^3 - 9x^2 - 24x - 4 = 0$
(B) $x^3 - 9x^2 - 12x - 4 = 0$
(C) $x^3 - 9x^2 - 24x - 16 = 0$
(D) $x^3 - 9x^2 - 12x - 16 = 0$

- 5 A particle of mass m is moving in a straight line under the action of a force.

$$F = \frac{m}{x^3}(6 - 10x)$$

What of the following is an expression for its velocity in any position, if the particle starts from rest at $x = 1$?

- (A) $v = \pm \frac{1}{x} \sqrt{(-3 + 10x - 7x^2)}$
(B) $v = \pm x \sqrt{(-3 + 10x - 7x^2)}$
(C) $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$
(D) $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x + 7x^2)}$

- 6 Which of the following is an expression for $\int \frac{2}{x^2 + 4x + 13} dx$?

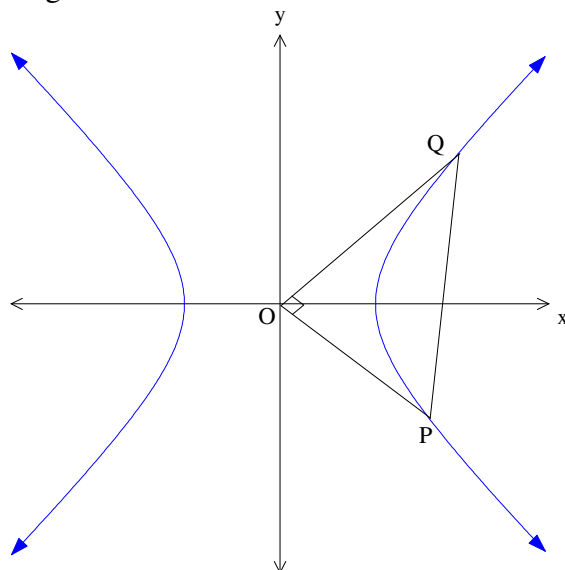
- (A) $\frac{1}{3} \tan^{-1} \frac{(x+2)}{3} + c$
(B) $\frac{2}{3} \tan^{-1} \frac{(x+2)}{3} + c$
(C) $\frac{1}{9} \tan^{-1} \frac{(x+2)}{9} + c$
(D) $\frac{2}{9} \tan^{-1} \frac{(x+2)}{9} + c$

- 7 Consider the hyperbola with the equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

What are the coordinates of the foci of the hyperbola?

- (A) $(\pm 4, 0)$
(B) $(0, \pm 4)$
(C) $(0, \pm 5)$
(D) $(\pm 5, 0)$

- 8 The diagram below shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a > b > 0$. The points $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \alpha, b \tan \alpha)$ lie on the hyperbola and the chord PQ subtends a right angle at the origin.



Use the parametric representation of the hyperbola to determine which of the following expressions is correct?

- (A) $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$
- (B) $\sin \theta \sin \alpha = \frac{a^2}{b^2}$
- (C) $\tan \theta \tan \alpha = -\frac{a^2}{b^2}$
- (D) $\tan \theta \tan \alpha = \frac{a^2}{b^2}$
- 9 It is given that $3 + i$ is a root of $P(z) = z^3 + az^2 + bz + 10$ where a and b are real numbers. Which expression factorises $P(z)$ over the real numbers?
- (A) $(z - 1)(z^2 + 6z - 10)$
- (B) $(z - 1)(z^2 - 6z - 10)$
- (C) $(z + 1)(z^2 + 6z + 10)$
- (D) $(z + 1)(z^2 - 6z + 10)$

10 If $x^3 + y^3x = y^2$, then $\frac{dy}{dx}$ is given by:

(A) $\frac{3x^2 + y^3}{2y - 3y^2x}$

(B) $\frac{3x^2 + y^3}{3y^2x - 2y}$

(C) $\frac{3x^2 + 3y^2x + y^3}{2y}$

(D) $\frac{3x^2 + y^3}{3y^2x - 2y}$

End of Section I

Section II

90 Marks

Attempt Questions 11-16

Allow about 2 hours and 40 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available

All necessary working should be shown in every question.

Question 11 (15 marks) Use a SEPARATE writing booklet	Marks
(a) If $z = 4 - 2i$ and $w = 3 + i$, evaluate $z^2 + \bar{w}$.	2
(b) (i) Find the Cartesian equation of the locus of z if $\arg\left(\frac{z-2}{z}\right) = \frac{\pi}{2}$.	1
(ii) Sketch the locus from part (i)	1
(c) Find $\int \frac{\sin x}{\cos^3 x} dx$	2
(d) (i) Express $-\sqrt{3} - i$ in modulus argument form.	2
(ii) Show that $(-\sqrt{3} - i)^6$ is a real number	2
(e) By considering the function $f(x) = x^2 - 4x $ sketch the graph of $y = \frac{1}{f(x)}$	2
(f) Find $\int \frac{1}{x^2 \sqrt{x^2 - 4}} dx$	3

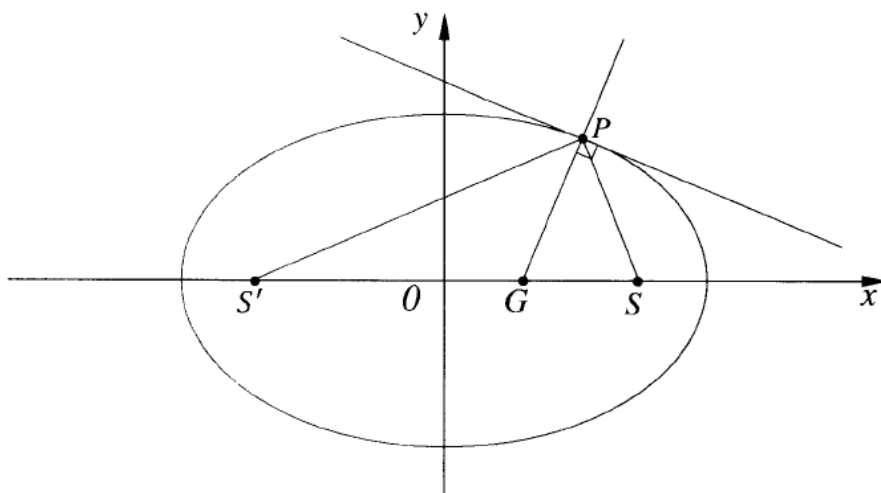
End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet**Marks**

(a) By using the substitution $t = \tan\left(\frac{x}{2}\right)$, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x}$.

4

- (b) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has foci $S(ae, 0)$ and $S'(-ae, 0)$, and directrices $x = \pm \frac{a}{e}$.
 $P(a\cos\theta, b\sin\theta)$ is a point on the ellipse with the normal at P meeting the x -axis at G .



- (i) Using the focus/directrix definition of an ellipse show that

2

$$\frac{PS}{PS'} = \frac{1 - e \cos \theta}{1 + e \cos \theta}.$$

- (ii) The equation of the normal at P is given by

3

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2. \text{ (Do NOT prove this.)}$$

Show that $\frac{GS}{GS'} = \frac{PS}{PS'}.$

(c) Let $I_n = \int_1^e x(\ln x)^n dx$ for $n = 0, 1, 2, \dots$, show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$

3

(d) Suppose that the complex number z lies on the unit circle, and $0 \leq \arg(z) \leq \frac{\pi}{2}$.

3

By the use of a suitable vector diagram, prove that $2 \arg(z+1) = \arg(z)$.

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) A particle of mass m is thrown vertically upwards with initial velocity U in a medium with resistive force $R = mkv$ where v is the velocity of the particle at time t and k is a constant.

The equation of the motion of the particle is then $\frac{dv}{dt} = -g - kv$ where g is the acceleration due to gravity (**Do not** prove this).

- (i) Use $\frac{dv}{dt} = v \frac{dv}{dx}$ to show that the vertical displacement x from the point of projection of the particle is given by

3

$$x = \frac{1}{k}(U - v) - \frac{g}{k^2} \log_e \left(\frac{g + kU}{g + kv} \right).$$

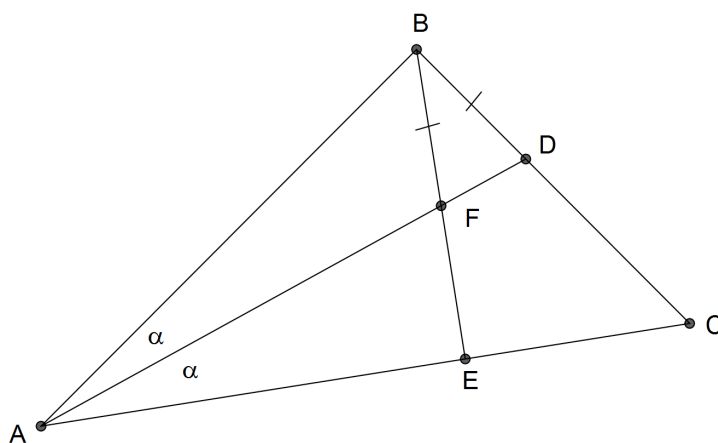
- (ii) Hence find an expression for H the maximum height reached by the particle.

1

- (iii) Find an expression for the time taken for the particle to reach its maximum height.

3

- (b) In the diagram below, AD bisects $\angle BAC$ and F is the point on AD so that $BF = BD$. Prove that AB is tangential to the circle passing through B , C and E .

3

Question 13 is continued on the next page

Question 13 continued**Marks**

- (c) The hyperbola $\mathcal{H}$ has equation $xy = 16$. The points $P\left(4p, \frac{4}{p}\right)$ for $p > 0$ and $Q\left(4q, \frac{4}{q}\right)$ for $q > 0$ are two distinct arbitrary points on $\mathcal{H}$.

(i) Show that the equation of the tangent at P is $x + p^2y = 8p$ **1**

(ii) Find the coordinates of T , the point of intersection of the tangents at P and Q . **2**

(iii) The equation of the chord passing through PQ is given by $px + y = 4(p + q)$ (**Do not** prove this).

If chord PQ passes through the point $N(0, 8)$ find the Cartesian equation of the locus of T **2**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

Marks

(a) Find $\int \frac{x^2 - 2x - 3}{(x+2)(x^2+1)} dx$

3

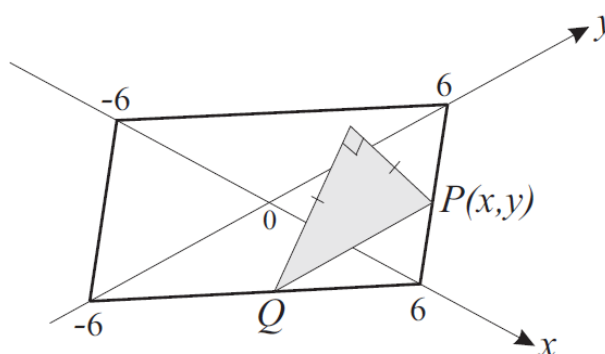
(b) (i) Draw a one third page sketch the graph of $y = \frac{x^3}{x^2 - 4}$,
indicating the coordinates of all stationary points and all asymptotes.

4

(ii) For what values of k will $x^3 - kx^2 + 4k = 0$ have exactly one real root.

1

(c)



The diagram above shows the horizontal square base of a solid. Vertical cross-sections of the solid perpendicular to the x -axis are right-angled isosceles triangles with hypotenuse in the base.

(i) Find, as a function of x , the area of a typical cross-section standing on the interval PQ .

2

(ii) Find the volume of the solid.

2

(d) If $U_1 = 8$ and $U_2 = 20$ and $U_n = 4U_{n-1} - 4U_{n-2}$ for $n \geq 3$,
prove by mathematical induction that $U_n = (n+3)2^n$ for $n \geq 1$

3

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet

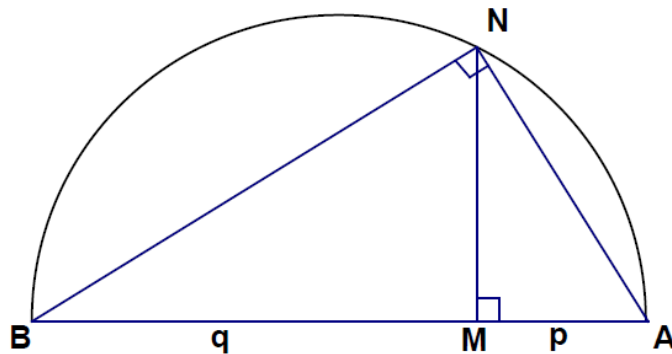
Marks

- (a) Show by the use of calculus that $x \geq \ln(x+1)$ for $x > -1$

Hint: Let $f(x) = x - \ln(x+1)$.

3

- (b) In the diagram, AB is the diameter of a semicircle. $\angle ANB = 90^\circ$ and M is a point on AB such that NM is perpendicular to AB .



If $AM=p$ and $BM=q$.

- (i) Explain why $NM = \sqrt{pq}$

1

- (ii) By reference to the geometry of the diagram deduce that $\sqrt{pq} \leq \frac{p+q}{2}$

1

- (iii) Hence prove that for $p, q, x, y \geq 0$ then

$$\frac{1}{4}(p+q+x+y) \geq (pqxy)^{\frac{1}{4}}$$

2

- (iv) Deduce that if $k, l, m, n \geq 0$ then $\frac{k}{l} + \frac{l}{m} + \frac{m}{n} + \frac{n}{k} \geq 4$

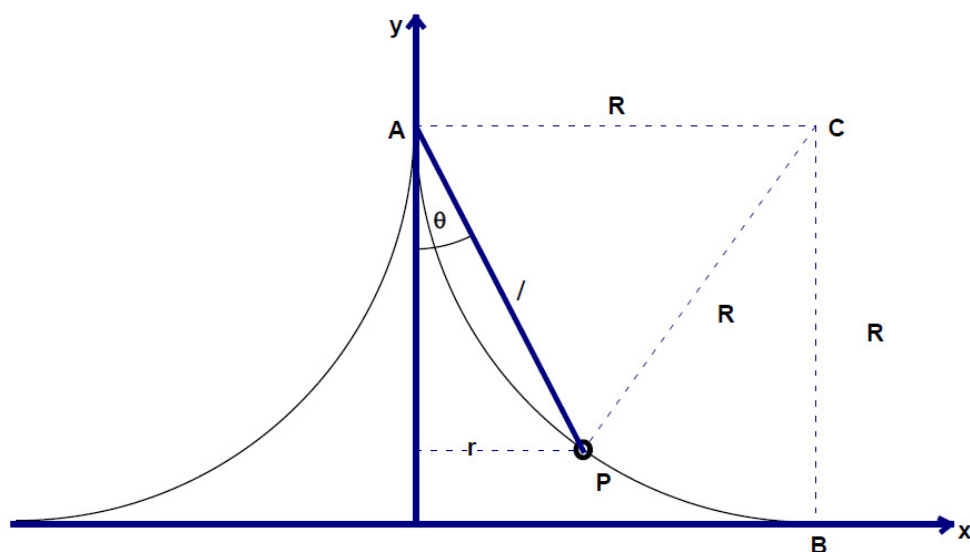
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Question 15 is continued on the next page

Question 15 continued

Marks

(c)



AB is an arc of a circle centre C and radius R . A surface is formed by rotating the arc AB through one revolution about the y -axis. A light, inextensible string of length l , $l \leq R$, is attached to point A , and a particle of mass m is attached to the other end. The particle is set in motion, tracing out a horizontal circle on the surface with constant angular velocity ω radians per second, while the string stays taut.

i) Explain why, when the particle is in position P shown on the diagram, the direction of the force N exerted by the surface on the particle is towards C . 1

ii) If the string makes an angle θ with the vertical, show that $\angle ACP = 2\theta$. 1

iii) Show on a diagram the tension force T , the force N and the weight force of magnitude mg acting on the particle, indicating their direction in terms of θ . 1

iv) Show that 2

$$T \cos \theta + N \sin 2\theta = mg$$

$$T \sin \theta - N \cos 2\theta = m l \sin \theta \omega^2$$

v) Show that 1

$$N = m l \sin \theta \left(\frac{g}{l} \sec \theta - \omega^2 \right).$$

vi) Deduce that there is a maximum value ω for the motion to occur as described, and write down this maximum value. 1

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) A bag contains 10 black and 10 blue marbles. Six marbles are selected without replacement.
- (i) Calculate the probability that exactly three marbles selected are blue, giving your answer correct to three decimal places. **1**
- (ii) Hence, or otherwise, calculate the probability that more than three of the marbles selected are blue, giving your answer correct to three decimal places. **2**
- (b) (i) Find an expression for the limiting sum of infinite geometric series $1 + z + z^2 + \dots$ for $|z| < 1$ **1**
- (ii) Given that complex number $z = \frac{1}{2}(\cos \theta + i \sin \theta)$, use your answer in part (i) to show that the imaginary part of $1 + z + z^2 + \dots$ is $\frac{2 \sin \theta}{5 - 4 \cos \theta}$. **3**
- (iii) Find an expression for $1 + \frac{1}{2} \cos \theta + \frac{1}{2^2} \cos 2\theta + \frac{1}{2^3} \cos 3\theta + \dots$ in terms of $\cos \theta$ **2**
- (c) (i) Find $\lim_{n \rightarrow \infty} [\tan^{-1}(n+1) + \tan^{-1}(n)]$. **1**
- (ii) Show that $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right)$, where n is a positive integer. **2**
- (iii) Hence show that $\lim_{n \rightarrow \infty} \sum_{j=1}^n \tan^{-1}\left(\frac{2}{j^2}\right) = \frac{3\pi}{4}$ **3**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

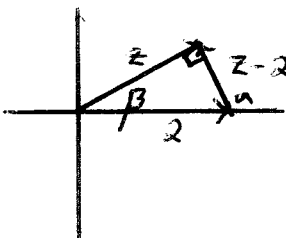
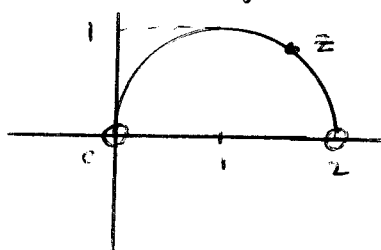
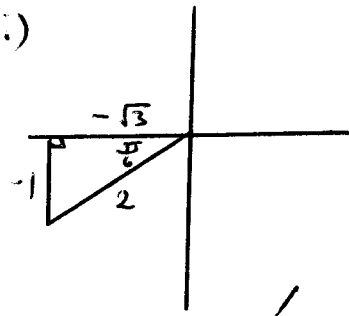
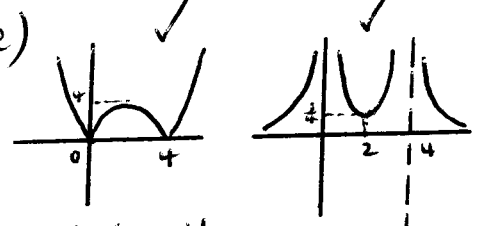
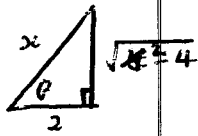


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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
1. B 2. D 3. A 4. C 5. C 6. B 7. D 8. A 9. D 10. A			
<u>Section I (Solutions)</u>			
1. $\frac{5}{iW} = \frac{5}{i(-2+i)}$ $= \frac{5}{-1-2i} \times \frac{-1+2i}{-1+2i}$ $= \frac{-1+2i}{-1-2i} \therefore B$		6. $\int \frac{2}{x^2+4x+13} dx$ $= \int \frac{2}{x^2+4x+4+9} dx$ $= \int \frac{2}{(x+2)^2+3^2} dx$ $= \frac{2}{3} \tan^{-1}\left(\frac{x+2}{3}\right) + C \therefore B$	
2. $\Delta V = 2\pi xy \cdot Ax$ $y = \sqrt{30-x^2} - x^2$ $V = \int_0^{\sqrt{30}} 2\pi x (\sqrt{30-x^2} - x^2) dx$ $\therefore D$		7. $a=4, b=3 \quad b^2 = e^2(e^2-1)$ $\therefore e^2 = \frac{25}{16} \quad e = \frac{5}{4}$ $\therefore S(\pm ai, 0) = S(\pm 5, 0) \therefore D$	
3. By inspection since $f(x) \geq 0 \therefore A$		8. $m_{OA} \times m_{OP} = -1$ $\frac{b \tan \alpha}{a \sec \alpha} \times \frac{b \tan \theta}{a \sec \theta} = -1$ $\therefore \sin \alpha \sin \theta = -\frac{a^2}{b^2} \therefore A$	
4. Let $y = x^{\frac{1}{2}}$ $y^{\frac{3}{2}} + 3y = -4$ $\Rightarrow y^{\frac{3}{2}} = -3y - 4$ expanding on squaring gives $y^3 - 9y^2 - 24y - 16 = 0$ $\therefore C$		9. Since $3+i$ is a root $\therefore 3-i$ also a root ($P(z)$ has real co-efficients). $\therefore z^2 - 6z + 10$ is a factor and $\Sigma \text{ roots} = -10 \therefore D$	
5. $\frac{dx}{x^2} = \frac{6}{x^3} - \frac{10}{x^2}$ $\frac{1}{2}v^2 = -3x^2 + 10x^{-1} + C$ at $v=0, x=1 \therefore C = -7$ $\therefore v^2 = \frac{2}{x^2} (-3+10x-7x^2)$ $\therefore v = \pm \frac{1}{x} \sqrt{-3+10x-7x^2} \therefore C$		10. $x^3 + y^3 x = y^2$ $3x^2 + y^3 + x \cdot 3y^2 \cdot \frac{dy}{dx} = 2y \cdot \frac{dy}{dx}$ $\therefore \frac{dy}{dx} = \frac{3x^2 + y^3}{2y - 3xy^2} \therefore A$	



2013 Year 12 Mathematics Extension 2 Trial HSC SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 11</u> (a) $z^2 + \bar{w} = (4-2i)^2 + 3-i$ $= 12 - 16i + 3 - i$ $= 15 - 17i$ (b) $\arg(z-2) - \arg(z) = \frac{\pi}{2}$ let $\alpha - \beta = \frac{\pi}{2}$ $\therefore \alpha = \beta + \frac{\pi}{2}$  i) So locus is semi circle centre (1, 0), radius 1 unit $y = \sqrt{1 - (x-1)^2}$ $y = \sqrt{2x - x^2}, 0 \leq y \leq 1$ $\text{or } (x-1)^2 + y^2 = 1, 0 \leq y \leq 1$ ii)  c) $\int \sin x (\cos x)^{-3} dx$ $= \int -\sin x (\cos x)^{-3} dx$ $= -\frac{1}{2} (\cos x)^{-2} + C$ $= -\frac{1}{2} \sec^2 x + C$	✓ ✓	d) :  $-\sqrt{3} - i = 2 \operatorname{cis} \left(\frac{7\pi}{6} \right)$ or equivalent. ii) $(-\sqrt{3} - i)^6 = \left[2 \operatorname{cis} \left(\frac{7\pi}{6} \right) \right]^6$ $= 2^6 \operatorname{cis} (7\pi)$ by DMT ✓ $= 2^6 \times -1$ $= -64 \therefore \text{Real}$ ✓ e)  $y = x(x-4)$ f) let $x = 2 \sec \theta \therefore dx = 2 \sec \theta \tan \theta d\theta$ $\therefore \int \frac{1}{4 \sec^2 \theta \sqrt{4 \sec^2 \theta - 4}} \cdot 2 \sec \theta \tan \theta d\theta$ $= \frac{1}{2} \int \frac{\tan \theta}{\sec \theta \sqrt{4 \sec^2 \theta - 4}} d\theta$ $= \frac{1}{4} \int \cos \theta d\theta$ ✓ $= \frac{1}{4} \sin \theta + C$ $= \frac{\sqrt{x^2 - 4}}{4x} + C$ ✓ 	✓ ✓

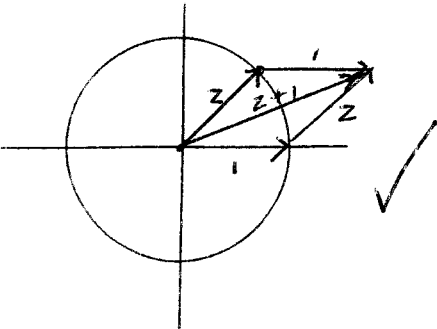


2013 Year 12 Mathematics Extension 2 Trial HSC SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 12</u> Let $I = \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} \cdot dx$ and let $t = \tan\left(\frac{x}{2}\right)$ $\therefore dx = \frac{2dt}{1+t^2}$ and $\left. \begin{matrix} x=0, & t=0 \\ x=\frac{\pi}{2}, & t=1 \end{matrix} \right\} \checkmark$ $\therefore I = \int_0^1 \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2} \checkmark$ $= \int_0^1 \frac{2}{t^2 + 2t + 1} dt \checkmark$ $= \int_0^1 2(t+1)^{-2} dt$ $= -2[(t+1)^{-1}]_0^1$ $= -2\left(\frac{1}{2} - 1\right)$ $= 1 \checkmark$ b)i) Let M and M' be the feet of the perpendiculars from P to the directrices $x = \frac{a}{e}$ and $x = -\frac{a}{e}$ Since $PS = ePM$ $= e\left(\frac{a}{e} - a\cos\theta\right)$ $= a(1 - e\cos\theta) \checkmark$		and $PS' = ePM'$ $= e\left(a\cos\theta + \frac{a}{e}\right)$ $= a(1 + e\cos\theta) \checkmark$ $\therefore \frac{PS}{PS'} = \frac{1 - e\cos\theta}{1 + e\cos\theta}$ ii) Normal at P meets x-axis at G. at $y=0$, $x = \frac{(a^2 - b^2)\cos\theta}{a}$ and since $b^2 = a^2(1 - e^2)$ $x = \frac{a^2 e^2 \cos\theta}{a}$ $= ae^2 \cos\theta \checkmark$ $\therefore G(ae^2 \cos\theta, 0)$ with $S(ae, 0)$ & $S'(-ae, 0)$ $\therefore GS = ae - ae^2 \cos\theta$ and $GS' = ae + ae^2 \cos\theta$ So $\frac{GS}{GS'} = \frac{ae - ae^2 \cos\theta}{ae + ae^2 \cos\theta}$ $= \frac{1 - e\cos\theta}{1 + e\cos\theta}$ $= \frac{PS}{PS'}$ as required c) $I_n = \int_1^e x(\ln x)^n dx$ $u = (\ln x)^n$ $= \frac{x^2(\ln x)^n}{2} \Big _1^e - \frac{1}{2} \int_1^e x(\ln x)^{n-1} dx$ $u' = \frac{n(\ln x)^{n-1}}{x} \checkmark$ $= \frac{e^2(\ln e)^n}{2} - 0 - \frac{1}{2} n \int_1^e x(\ln x)^{n-1} dx$ $v' = x \checkmark$ $= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ $v = \frac{x^2}{2} \checkmark$	



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d)  let $\arg(z) = \theta$ $\arg(z+1) = \frac{\theta}{2}$ (diagonal of rhombus) ✓ bisects the angle So $2\arg(z+1) = \theta$ ✓ and equating $2\arg(z+1) = \arg(z)$		$= -\frac{1}{kv} \int 1 dv + \frac{g}{k^2} \int \frac{kv}{g+kv} dv$ $= -\frac{v}{kv} + \frac{g}{k^2} \ln(g+kv) + C \quad \checkmark$ at $x=0 \quad v=U$ $\therefore C = \frac{U}{kv} - \frac{g}{k^2} \ln(g+kU) \quad \checkmark$ $\therefore x = \frac{U}{k} - \frac{v}{kv} - \frac{g}{k^2} [\ln(g+kU) - \ln(g+kv)]$ $\therefore x = \frac{1}{k} (U-v) - \frac{g}{k^2} \left[\ln\left(\frac{g+kU}{g+kv}\right) \right]$ as required. ii) Max Height $x=H$ when $v=0$ $\therefore x = \frac{1}{k} U - \frac{g}{k^2} \left[\ln\left(\frac{g+kU}{g}\right) \right] \quad \checkmark$ iii) $\frac{dv}{dt} = -g - kv$ $\therefore \frac{dt}{dv} = \frac{-1}{g+kv}$ $\therefore t = \int \frac{-1}{g+kv} dv$ $= -\frac{1}{k} \int \frac{kv}{g+kv} dv$ so $t = -\frac{1}{k} \ln(g+kv) + C$ at $t=0, v=U \therefore C = \frac{1}{k} \ln(g+kU)$ $\therefore t = \frac{1}{k} \ln\left(\frac{g+kU}{g+kv}\right) \quad \checkmark$ at Max height $v=0$ $\therefore T = \frac{1}{k} \ln\left(\frac{g+kU}{g}\right) \quad \checkmark$	
<u>Question 13</u> a) i) let $v \frac{dv}{dx} = -g - kv$ $\frac{dx}{dv} = \frac{-v}{g+kv}$ so $x = \int \frac{-v}{g+kv} dv \quad \checkmark$ $= -\frac{1}{k} \int \frac{g+kv-g}{g+kv} dv$ $= -\frac{1}{k} \int 1 - \frac{g}{g+kv} dv$			



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b) For AB to be tangential to circle thro BCE, need to show $\angle ABE = \angle BCE$. Let $\angle ABF = \beta$ $\therefore \angle BFD = \alpha + \beta$ (Ext $\angle$ of $\triangle ABF$) $= \angle BDF$ ($BF = BD$ given) $\therefore \angle FBD = \pi - (2\alpha + 2\beta)$ ($\angle$ sum of $\triangle$) ✓ Also $\angle AFE = \alpha + \beta$ (Vert. Opp). $\therefore \angle FED = 2\alpha + \beta$ (Ext. $\angle$ of $\triangle FAE$) $\therefore \angle BCE = \pi - (\pi - (2\alpha + 2\beta)) - (\alpha + \beta)$ $= \pi - \pi + 2\alpha + 2\beta - \alpha - \beta$ $= \beta$ ✓ $\therefore \angle ABE = \angle BCE$ so AB is tangential as angle made between tangent and chord BE is equal to angle in the alternate segment. ✓		c) i) $xy = 16$ $\frac{d}{dx}(xy) = 0$ $x \frac{dy}{dx} + y = 0$ $\therefore \frac{dy}{dx} = -\frac{y}{x}$ at R $m_T = \frac{-\frac{4}{p}}{\frac{4p}{p^2}}$ $= -\frac{1}{p^2}$ $\therefore y - \frac{4}{p} = -\frac{1}{p^2}(x - 4p)$ $p^2y - 4p = -x + 4p$ ✓ $\therefore x + p^2y = 8p$ ii) Similarly $x + q^2y = 8q$ for tangent at Q. subtracting. $(p^2 - q^2)y = 8(p - q)$ $\therefore y = \frac{8}{p+q}$ ✓ so $x + \frac{8p^2}{p+q} = 8p$ $x = \frac{8p^2 + 8pq - 8p^2}{p+q}$ $= \frac{8pq}{p+q}$ ✓ $\therefore T\left(\frac{8pq}{p+q}, \frac{8}{p+q}\right)$	

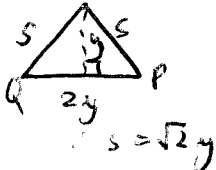


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iii) $N(0,8) \therefore p+q=2$ for T $x = \frac{8pq}{p+q}$ and $y = \frac{8}{p+q}$ so $y=4$ ✓ but if $p, q > 0$ then $x = \frac{8pq}{p+q} > 0$ and since $p+q=2$ $pq=2p-p^2$ has max value at 1 $\therefore$ locus is $y=4, 0 < x < 4$ ✓		b) i) $y = \frac{x^3}{x^2-4}$ $x^2-4 \overline{) x^3+0x^2+0x+0}$ $\underline{x^3-4x}$ $+4x$ $= x + \frac{4x}{x^2-4} = x + \frac{4x}{(x-2)(x+2)}$ $y' = \frac{(x^2-4) \cdot 3x^2 - x^3(2x)}{(x^2-4)^2}$ $= \frac{3x^4 - 12x^2 - 2x^4}{(x^2-4)^2}$ $= \frac{x^4 - 12x^2}{(x^2-4)^2} = 0$ wh $x^2=0 \Rightarrow x=0$ $x^2=12 \Rightarrow x = \pm 2\sqrt{3}$ $y = \frac{(2\sqrt{3})^3}{8} = \frac{8 \times 5\sqrt{3}}{8} = 3\sqrt{3}$ $\therefore$ Stat pts $(\pm 2\sqrt{3}, 3\sqrt{3})$ (0,0) Asymptotes: $x \neq \pm 2$ as $x \rightarrow \infty, y \rightarrow x^+$ Odd function.	
<u>Question 14</u> a) let $\frac{x^2-2x-3}{(x+2)(x^2+1)} \equiv \frac{a}{x+2} + \frac{bx+c}{x^2+1}$ $x^2-2x-3 \equiv a(x^2+1) + (bx+c)(x+2)$ let $x=-2 \Rightarrow 5 = 5a \Rightarrow a=1$ let $x=0 \Rightarrow -3 = 1+2c \Rightarrow c=-2$ let $x=1 \Rightarrow -4 = 2 + (b-2)3$ $-6 = 3b-6$ $b=0$ $\therefore$ let $I = \int \frac{1}{x+2} dx + \int \frac{-2}{x^2+1} dx$ $= \ln x+2 - 2 \tan^{-1}(x) + C$ ✓			



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ii) Consider when $\frac{x^3}{x^2-4} = k \text{ has one soln.}$ by inspection of graph: $-3\sqrt{3} < k < 3\sqrt{3}$ c) i)  $s = \sqrt{2}y$ $\text{Area} = \frac{1}{2} \times 6y \times y$ $= 3y^2$ $PQ = (6-x) - (x-6)$ $= 12 - 2x$ $= 2(6-x)$ $\therefore \text{Area} = (6-x)^2$ ii). $V = 2 \int_0^6 (6-x)^2 dx \quad (\text{symmetry of solid})$ $= 2 \left[\frac{(6-x)^3}{-3} \right]_0^6$ $= 2 \left(0 - \left(\frac{6^3}{-3} \right) \right)$ $= 144 \text{ u}^3$		d) When $n=1$, $U_1 = 4 \times 2^1 = 8$ and $n=2$ $U_2 = 5 \times 2^2 = 20$ $\therefore$ Statement True for $n=1, 2$ Assume statement true for $n=k, n=k-1$ $\therefore U_{k-1} = (k+2) \cdot 2^{k-1}$ $U_k = (k+3) \cdot 2^k$ When $n = k+1$ $U_n = U_{k+1}$ $= 4U_{(k+1)-1} - 4U_{(k+1)-2}$ $= 4U_k - 4U_{k-1}$ $= 4(k+3)2^k - 4(k+2)2^{k-1}$ $= (4k+12) \cdot 2^k - 2(k+2) \cdot 2^k$ $= 2k \cdot 2^k - 8 \cdot 2^k$ $= k \cdot 2^{k+1} + 4 \cdot 2^{k+1}$ $= (k+4) \cdot 2^{k+1}$ $= (n+3) \cdot 2^n \text{ True}$ $n = k+1$	✓ ✓ ✓



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<u>Question 15</u> (a) $f(x) = x - \ln(x+1)$ $f'(x) = 1 - \frac{1}{x+1}$ $f''(x) = \frac{1}{(x+1)^2}$ ✓ Noting $f'(0) = 0$ $f''(0) > 0$ $(0,0)$ is a min. stat. pt. ✓ Noting $f''(x) > 0$ for all $x > -1$ $f(x)$ is concave up for all $x > -1$ ✓ $f(x) > 0$ $\therefore x > \ln(1+x)$ (b) i) Since $\triangle BMN \parallel \triangle NMA$ $\frac{BM}{NM} = \frac{NM}{MA}$ ✓ $NM^2 = AM \cdot MB$ $= p \cdot q$ $\therefore NM = \sqrt{pq}$ ii) $AB = p+q$ is diameter $\therefore \frac{p+q}{2}$ is radius Since $NM \perp AB$, MN is always less than length of radius $NM \leq \frac{p+q}{2}$ $\therefore \sqrt{pq} \leq \frac{p+q}{2}$		iii) $\frac{1}{4}(p+q+x+y)$ $= \frac{1}{2}\left(\frac{p+q}{2} + \frac{x+y}{2}\right)$ ✓ $\geq \frac{1}{2}(\sqrt{pq} + \sqrt{xy})$ $\geq \sqrt{\sqrt{pq} \cdot \sqrt{xy}}$ ✓ $\geq (pqxy)^{\frac{1}{4}}$ iv) Replacing p, q, x, y with $\frac{k}{l}, \frac{l}{m}, \frac{m}{n}, \frac{n}{k}$ ✓ $\frac{1}{4}\left(\frac{k}{l} + \frac{l}{m} + \frac{m}{n} + \frac{n}{k}\right) \geq \left(\frac{k}{l} \cdot \frac{l}{m} \cdot \frac{m}{n} \cdot \frac{n}{k}\right)^{\frac{1}{4}}$ $\therefore \frac{k}{l} + \frac{l}{m} + \frac{m}{n} + \frac{n}{k} \geq 4$ (c) (i) The normal force is at right angles to the tangent to the circle and is therefore directed towards the circle's centre C. ✓ ii) In $\triangle ACP$, $\angle PAC = \frac{\pi}{2} - \theta$ (complementary, y-axis tangent to circle). $= \angle APC$ ($AC=AP$, equal radii) $\therefore \angle ACP = \pi - 2\left(\frac{\pi}{2} - \theta\right)$ ($\angle$ sum of $\triangle$) ✓ $= 2\theta$	



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iii) iv) Vertically forces are balanced ✓ $T \cos \theta + N \sin 2\theta = mg \quad \text{--- (1)}$ Sum of radial forces = $m r \omega^2$ towards centre of motion. $T \sin \theta - N \cos 2\theta = m r \omega^2 \quad \checkmark$ but $\sin \theta = \frac{r}{l}$ $\therefore r = l \sin \theta$ $T \sin \theta = m l \sin \theta \omega^2 \quad \text{--- (2)}$ v) ① $\times \sin \theta$ - ② $\times \cos \theta$ gives $N \sin 2\theta \sin \theta + N \cos 2\theta \cos \theta = m g \sin \theta - m l \sin \theta \cos \theta \omega^2$ $N \cos (2\theta - \theta) = m \sin \theta (g - m l \cos \theta \omega^2) \quad \checkmark$ $\therefore N \cos \theta = m \sin \theta \cos \theta \left(\frac{g \sec \theta}{l} - \omega^2 \right)$ $\therefore N = m l \sin \theta \left(\frac{g}{l} \sec \theta - \omega^2 \right)$		vi) For the motion $N \geq 0$ $\therefore \frac{g \sec \theta}{l} - \omega^2 \geq 0$ $\omega \leq \sqrt{\frac{g \sec \theta}{l}}$ $\therefore \text{Max is } \sqrt{\frac{g \sec \theta}{l}} \quad \checkmark$ <u>Question 16</u> (a) i) $P(3 \text{ blue}) = \frac{{}^{10}C_3 \times {}^{10}C_3}{{}^{20}C_6}$ $= 0.372 \text{ (3 d.p.)}$ ii) $P(> 3 \text{ blue})$ $= P(4 \text{ blue}) + P(5 \text{ blue}) + P(6 \text{ blue})$ $= \frac{1 - P(3 \text{ blue})}{2} \text{ (symmetry)}$ $= 0.314 \text{ (3 d.p.)} \quad \checkmark$ (b) i) $S_{\infty} = \frac{1}{1 - z} \quad \checkmark$	



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<u>Question 16 (continued)</u> ii) $\frac{1}{1-z} = \frac{1}{1 - (\frac{1}{2}\cos\theta + \frac{1}{2}i\sin\theta)}$ $= \frac{1}{(1 - \frac{1}{2}\cos\theta) - \frac{1}{2}i\sin\theta} \times \frac{1 - \frac{1}{2}\cos\theta + \frac{1}{2}i\sin\theta}{1 - \frac{1}{2}\cos\theta + \frac{1}{2}i\sin\theta}$ $= \frac{1 - \frac{1}{2}\cos\theta + \frac{1}{2}i\sin\theta}{(1 - \frac{1}{2}\cos\theta)^2 + \frac{1}{4}\sin^2\theta}$ $= \frac{1 - \frac{1}{2}\cos\theta + \frac{1}{2}i\sin\theta}{1 - \cos\theta + \frac{1}{4}\cos^2\theta + \frac{1}{4}\sin^2\theta}$ $= \frac{1 - \frac{1}{2}\cos\theta + \frac{1}{2}i\sin\theta}{\frac{5}{4} - \cos\theta} \checkmark$ $\therefore \operatorname{Im}\left(\frac{1}{1-z}\right) = \frac{\frac{1}{2}\sin\theta}{\frac{5}{4} - \cos\theta} \times \frac{4}{4}$ $= \frac{2\sin\theta}{5 - 4\cos\theta}$ iii) $1 + \frac{1}{2}\cos\theta + \frac{1}{2^2}\cos 2\theta + \frac{1}{3^2}\cos 3\theta + \dots$ is real part of $\frac{1}{1-z}$ by DMT. $\therefore \operatorname{Re}\left(\frac{1}{1-z}\right) = \frac{1 - \frac{1}{2}\cos\theta}{\frac{5}{4} - \cos\theta} \times \frac{4}{4}$ $= \frac{4 - 2\cos\theta}{5 - 4\cos\theta} \checkmark$		(c) i) π $\checkmark$ ii) Let $\alpha = \tan^{-1}(n+1)$ $\alpha \quad \beta = \tan^{-1}(n-1)$ $\therefore \tan(\alpha - \beta) \checkmark$ $= \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$ $= \frac{(n+1) - (n-1)}{1 + (n-1)(n+1)} = \frac{2}{n^2} \checkmark$ $\therefore \alpha - \beta = \tan^{-1}\left(\frac{2}{n^2}\right)$ iii) $\sum_{j=1}^n \tan^{-1}\left(\frac{2}{j^2}\right)$ $= \tan^{-1}\left(\frac{2}{1^2}\right) + \tan^{-1}\left(\frac{2}{2^2}\right) + \tan^{-1}\left(\frac{2}{3^2}\right) + \dots + \tan^{-1}\left(\frac{2}{(n-1)^2}\right) + \tan^{-1}\left(\frac{2}{n^2}\right) \checkmark$ $= \tan^{-1}(2) - \tan^{-1}(0)$ $+ \tan^{-1}(3) - \tan^{-1}(1)$ $+ \tan^{-1}(4) - \tan^{-1}(2)$ $+ \tan^{-1}(5) - \tan^{-1}(3)$ $+ \dots$ $+ \tan^{-1}(n-1) - \tan^{-1}(n-3)$ $+ \tan^{-1}(n) - \tan^{-1}(n-2)$ $+ \tan^{-1}(n+1) - \tan^{-1}(n-1)$ $= -\tan^{-1}(0) - \tan^{-1}(1) + \tan^{-1}(n+1) + \tan^{-1}(n)$ $= 0 - \frac{\pi}{4} + \pi \text{ as } n \rightarrow \infty \text{ from part (i)} \checkmark$	

$$\lim_{n \rightarrow \infty} \sum_{j=1}^n \tan^{-1}\left(\frac{2}{j^2}\right) = \frac{3\pi}{4}$$